First-Order Smoothing-Based Deep Graph Network for Hyperspectral Image Classification

Yizhen Li, Yanwen Chong, Shaoming Pan, and Yun Ding

***Abstract*— Although graph convolutional network (GCN) has achieved remarkable success in hyperspectral image (HSI) classification, most existing GCN-based approaches have failed to realize a deep network structure due to the oversmoothing problem. This problem largely limits the expression ability and feature extraction ability of GCN and hampers GCN's capacity to model long-range relationships between samples in hyperspectral (HS) scenes. Moreover, there is a lack of theoretical analysis in those works that constructed deep GCN for HSI classification to illustrate how they overcome the oversmoothing problem. Aside from this, the characteristics and complexity of HSI are often neglected when constructing deep GCN models in HSI classification. To address these problems, a novel deep graph network based on first-order smoothing is proposed for HSI classification. Specifically, a local and global topologically consistent graph is constructed to thoroughly explore the union between fine pixel information and semantic superpixel information. Subsequently, a novel propagation procedure is proposed to address the oversmoothing problem. We creatively build a residual connection to the first layer to emphasize the feature information aggregated from the first-order neighborhood, which adds node features that have not yet become indistinguishable into deep layer, and at the same time, it can be considered as a correction to the original pixels affected by spectral variation in the input graph. Finally, we demonstrate how first-order smoothing-based deep graph network (FSDGN) can slow down the convergence rate of the oversmoothing problem by analyzing the propagation of FSDGN from the standpoint of the Laplacian spectral domain. In addition, the results of experiments performed on three benchmark datasets demonstrate its superiority over other state-of-the-art methods.**

***Index Terms*— Deep graph neural network, deep learning (DL), first-order smoothing, graph convolutional network (GCN), hyperspectral image (HSI) classification.**

I. Introduction

With the remarkable development of optical science and imaging technology in data acquisition, processing, and application, hyperspectral (HS) remote sensing technology has made great progress and attracts extensive attention [1], [2], [3]. The HS image (HSI), with a significant ascendancy of containing hundreds of narrow spectral bands and abundant information, has been widely applied in resource surveys, mineral detection, environmental monitoring, and so on. Most of these tasks are inseparable from HSI classification, which is designed to label each pixel according to the characteristics of samples, and thereupon it becomes the focus of the study on HSI [4].

Various approaches of HSI classification have been extensively explored over the past few decades. At the early stage, due to the natural advantage of HSI in spectral resolution, people sought to make full use of the rich spectral information in HSI. Some kernel-based methods such as support vector machines (SVMs) [5], kernel Fisher discriminant (KFD) analysis [6], and principal component analysis (PCA) [7] were proposed for HSI classification. However, these methods neglected the spatial associations between pixels, which is vital for improving the performance of HSI classifiers, and spatial–spectral-based methods were put forward. For instance, Markov random field (MRF) [8] was adopted to incorporate spatial information in the process of classification and achieved noticeable accuracy. Recently, an unsupervised DR method [9] called superpixelwise collaborative representation graph embedding (SPCRGE) has also been proposed for HSI classification. In addition, a novel superpixelwise adaptive approach named SpaSSA [10] was proposed for effective feature extraction in HSI. Besides, there are some studies that focused on the cross scene HSI classification. To address the spectral shift issue in this field, Liu et al. [11] proposed a spectral shift mitigation (SSM) method that includes amplitude shift mitigation (ASM) and adjacency effect mitigation (AEM). A novel discriminative cooperative alignment (DCA) [12] approach is proposed to cooperatively reduce the geometric and statistical shift. Nevertheless, the above-mentioned methods rely significantly on handcrafted features and prior expertise, which is so empirical that it is hard to generalize among different classes and datasets.

The artificial neural network has boomed in recent years, coinciding with the rise of deep learning (DL). DL, in contrast to conventional approaches, may automatically learn data patterns and feature representations, thus eliminating the need for complex feature engineering. Methods under the framework of DL have sprung up in the field of HSI classification [13], [14], [15]. Chen et al. [16], who introduced the DL technique for HSI classification first, focused on applying an autoencoder to extract data features. In [17], a recurrent neural network (RNN) with gated recurrent units (GRUs) is applied to HSI classification tasks. To tackle the

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lack of sufficient ground truth information to train the DL networks, Sun et al. [18] proposed a Gumbel-Softmax (GS) trick-enabled concrete autoencoder-based unsupervised band selection framework (CAE-UBS) for HSI. Among all these DL-based methods, the convolutional neural network (CNN) is the most promising method due to its exceptional ability to extract accurate local contextual features. In [19], a feature learning CNN (fl-CNN) for feature extraction and a classification CNN (c-CNN) were used for pixel classification. To make the best use of the rich spectral information, Sun et al. [20] developed a new CNN framework which combined the spectral and spatial information by considering the correlations between different bands. Besides, some special CNN structures have also been applied and improved in HSI classification. In [21], a model based on pyramidal bottleneck residual units was proposed for robust deep feature learning. Cheng et al. [22] proposed a deep high-order tensor convolutional sparse coding (CSC) model to train deep high-order filters and extract high-order features. Moreover, some recent studies have focused on the fusion of multisource data. For example, an information fusion network [23] named IP-CNN is proposed for integrating heterogeneous information and improving joint classification performance of HSI and light detection and ranging (LiDAR) data.

Despite the excellent performance displayed by the CNN-based methods, there are still some drawbacks. With a regular and identical kernel for every pixel in an image, the convolution operation in CNN cannot construct a flexible homogenous topological structure for HSI. Furthermore, since the convolution kernels aggregates features from pixels in a local area, it is hard for them to build long-range connections between homogenous pixels. By comparison, graph convolutional network (GCN) learns node embeddings through graph structure, and methods based on GCN have evolved. Qin et al. [24] initially proposed using GCN for HSI classification. Subsequently, Sha et al. [25] proposed an edge-conditioned graph neural network for HSI classification. To generate promising node representations, the dynamic adaptive sampling GCN (DASGCN) [26] was developed, which adaptively refined the receptive field and edge weight of each node during the convolution process. In addition, there are attempts to acquire an effective graph structure for HSI. Wan et al. [27] proposed a dynamic graph convolution and used a multiscale method to build an accurate graph and adequately exploit the spatial information. In [28], a spatial–spectral unified adaptive probability graph structure that can precisely represent node relationships was defined to improve classification accuracy. In addition, there have been advancements in the application of CNN training tools for GCN-based HSI classification. Mini-batch sampling was recommended for use in GCN to reduce computation [29]. Zhang et al. [30] introduced a spatial pooling operation for GCN that preserves significant features while reducing the size of input patches. In the field of cross-scene HSI classification, to better characterize the topological relationship alignment, a topological structure and semantic information transfer network (TSTnet) [31] was proposed, and Zhang et al. [32] developed a graph information aggregation cross-domain few-shot learning (Gia-CFSL) framework to solve the domain shift problem.

Although these existing GCN-based methods have shown their efficiency in HSI classification, they have a common weakness in that they are constrained to shallow, two-layer architectures. In principle, deeper versions of GCN models are more expressive than shallow ones, because it is easy to fit a shallow model by a deep model, but not the other way around. Moreover, the capacity of deep graph architectures is enhanced by their potential for excavating long-range homogenous neighbors. As a result, high-level advanced information is easier to learn in deeper layers. Nonetheless, the oversmoothing problem [33], which occurs when Laplacian smoothing is repeatedly applied to the input graph, restricts GCN from extending to deep structures. There are some related studies exploring deep GCN in the field of computer vision [34], [35], [36], [37], and Yang et al. [38] proposed a deep adaptive graph network for HSI classification based on the GCN via initial residual and identity mapping (GCNII) [37]. In spite of their success in overcoming the depth limitation of GCN, some problems still exist. Foremost, to build a deep GCN structure, the oversmoothing problem is the key obstacle to conquer. There is a need for an effective propagation mode that can prevent model degradation in deep layers and extract high-order beneficial feature information. Nevertheless, since this deep graph convolution model is proposed for HSI classification tasks, the characteristics of HSI should be taken into consideration when designing the propagation mechanism. Furthermore, although satisfactory results have been achieved through applying existing deep GCN models on HSI classification, the operating principle of deep GCN in combination with the characteristics of HSI has not been adequately explained, leading to an ambiguous theoretical framework.

To address the problems we have discussed, in this study, a first-order smoothing-based deep graph network (FSDGN) is proposed for obtaining high-order distinctive graph representations in HSI classification. In particular, we initially propose a global and local consistent graph construction (GLGC) module, which uses the adaptive global consistent graph structure [39] and enhances it by forming local topologically reliable relationships between pixels to ensure the credibility of local adjacency relationships. In addition, we propose a deep graph propagation method based on the feature information smoothed by the first-order neighbors. To construct a deep GCN model that is capable of preventing oversmoothing problems, we incorporate the local features in shallow layers into the graph representations of deep layers. In this process, the graph representation of the first layer is particularly emphasized since the graph representation undergoing one-step propagation has the same function of preventing oversmoothing as the input feature, and the node features after aggregating information from their nearest neighbors can be seen as a kind of correction of noise in the initial HSI. For this purpose, a graph construction method based on topological relationships can assure the validity of the first-order neighbors

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in turn. Finally, we investigate the FSDGN model in depth. We decouple the proposed method into two operations: partial self-loop and initial feature connection and demonstrate its ability to solve the oversmoothing problem using spectral analysis. In conclusion, the important contributions of this work are summarized as follows.

1) The first-order smoothing-based graph convolution layer is proposed to enable FSDGN to improve its capacity to express and extract features in a deep mode. This proposed method incorporates first-order neighborhood information in each propagation, which reduces the deviation of the initial feature caused by spectral variation, and serves as lower order information, which alleviates the oversmoothing problem when layers are stacked.
2) A thorough spectral analysis of the proposed FSDGN model is presented. Through a detailed theoretical demonstration, we describe how FSDGN slows down the convergence rate of the oversmoothing problem from two aspects and prove its effectiveness in deep layers.
3) Experiments were performed on three benchmark datasets, and the experimental results demonstrate that compared with other state-of-the-art models, FSDGN is more competitive and has stronger classification abilities.

The rest of this article is organized as follows. Section II introduces the fundamentals of this study and GCN. The proposed FSDGN model is described in Section III. The spectral analysis of FSDGN is provided in Section IV. Section V contains the experimental results and the corresponding analysis. Section VI concludes the study.

II. Preliminaries and Related Works

A. Notations

Given an HSI, it is represented by $\mathbf{O} \in \mathbb{R}^{h\times w\times c}$, where h denotes its height, w denotes its width, and c denotes the number of spectral bands. Denoted by $\mathbf{X} \in \mathbb{R}^{N\times c} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]^T$, the 3-D HSI cube $\mathbf{O}$ is transformed into the 2-D form, where $N = h * w$ and $\boldsymbol{x}_i \in \mathbb{R}^C$ is the spectral signature of the ith pixel. And the coordinates of pixels in $\mathbf{X}$ are denoted by $\mathbf{T} \in \mathbb{R}^{N\times 2} = [\boldsymbol{t}_1, \boldsymbol{t}_2, \ldots, \boldsymbol{t}_N]^T$.

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, $\mathcal{V}$ is the node set of $\mathcal{G}$ and $\mathcal{E}$ is the edge set. Denoted by $\mathbf{A}$, the adjacency matrix records the adjacent relationship of nodes, which is nonnegative, and a_{ij} denotes the adjacent relationship between node i and node j, which is positive if there is an edge between node i and node j. Denoted by $\mathbf{D}$, the degree matrix is a diagonal matrix, where d_{ii} denotes the degree of node i, and $d_{ii} = \sum_j a_{ij}$. Denoted by $\mathbf{L}$, the graph Laplacian is computed by $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and its normalized form $\mathbf{L}_{\text{sym}}$ is defined by $\mathbf{L}_{\text{sym}} = \mathbf{D}^{-(1/2)}\mathbf{L}\mathbf{D}^{-(1/2)} = \mathbf{I} - \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}$, where $\mathbf{I}$ is the identity matrix. $\mathbf{L}_{\text{sym}}$ can be further decomposed as $\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$ through eigendecomposition, where $\mathbf{U}$ is the matrix of eigenvectors, $\mathbf{\Lambda}$ is a diagonal matrix, and the diagonal elements are eigenvalues of $\mathbf{L}_{\text{sym}}$.

B. Basics of Graph Convolutional Network

Based on the Fourier transform, considering a signal $x \in \mathbb{R}^c$, the graph convolution on x is defined by a filter $g_\vartheta(\mathbf{\Lambda})$ [40], [41], which is a function of $\mathbf{\Lambda}$. We present it in the mathematical form as

$$g_\vartheta * x = \mathbf{U} g_\vartheta(\mathbf{\Lambda})\mathbf{U}^T x \tag{1}$$

where $\vartheta \in \mathbb{R}^n$ is the filter coefficient vector. The model of (1) requires high computation cost due to the multiplication with the Fourier basis $\mathbf{U}$. To solve this problem, the Kth order Chebyshev polynomials are applied to approximate $g_\vartheta(\mathbf{\Lambda})$ as a Chebyshev expansion that can be recursively computed [42], i.e.,

$$g_{\theta'} * x \approx \mathbf{U}\left(\sum_{k=0}^{K} \theta'_k T_k\left(\hat{\Lambda}\right)\right)\mathbf{U}^T x \tag{2}$$

where $\theta' \in \mathbb{R}^{K+1}$ is the polynomial coefficient vector, $\hat{\mathbf{\Lambda}}$ is constrained in $[-1, 1]$ via a scale operation of $\hat{\mathbf{\Lambda}} = (2/\lambda'_{\max})\mathbf{\Lambda} - \mathbf{I}$, with $\lambda'_{\max}$ being the largest eigenvalue of $\mathbf{L}_{\text{sym}}$, and $T_k(x)$ is defined as $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$, with $T_0(x) = 1$ and $T_1(x) = x$. Expand the Chebyshev polynomials in (2) and then we can obtain

$$g_\theta * x = \mathbf{U}\left(\sum_{k=0}^{K} \theta_k \mathbf{\Lambda}^k\right)\mathbf{U}^T x = \left(\sum_{k=0}^{K} \theta_k \mathbf{L}^k_{\text{sym}}\right)x \tag{3}$$

where $\theta \in \mathbb{R}^{K+1}$ is a coefficient vector, and θ_k is linearly combined by $\theta'_k (k = 0, 1, \ldots, K)$. Kipf and Welling [43] further add a limitation of $K = 1$ to (3), and set $\theta_0 = 2\theta$ and $\theta_1 = -\theta$. The model was simplified as

$$g_\theta * x = \theta\left(\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)x. \tag{4}$$

Ultimately, the *renormalization trick* is used to replace $\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}$ by $\widetilde{\mathbf{D}}^{-(1/2)}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-(1/2)}$, with $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\widetilde{\mathbf{D}} = \mathbf{D} + \mathbf{I}$. Finally, a nonlinear activation function is added, and the final representation of the graph convolution operation is

$$\mathbf{H}^{l+1} = \sigma\left(\hat{\mathbf{A}}\mathbf{H}^l\Theta_l\right) \tag{5}$$

where $\hat{\mathbf{A}} = \widetilde{\mathbf{D}}^{-(1/2)}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-(1/2)}$, Θ_l is the weight matrix of layer l, and σ is the activation function rectified linear unit (ReLU).

III. FSDGN Model

Fig. 1 shows the framework of our proposed FSDGN model. We show the whole transformation of HSI pixels into graph nodes and then participate in our first-order smoothing-based graph convolution. Using this architecture, we intend to extract high-order feature information through initial feature connection and partial self-loop operation. Specifically, the simple linear iterative clustering (SLIC) algorithm is initially applied to the HSI to create spatially correlated superpixels. Then, we use an improved local-adaptative and global-consistent graph construction method to construct a dependable HSI graph structure. Finally, this reliable graph structure is applied to several first-order smoothing-based graph convolution layers to uncover underlying deep information and provide discriminative graph representations, while preventing the oversmoothing problem.

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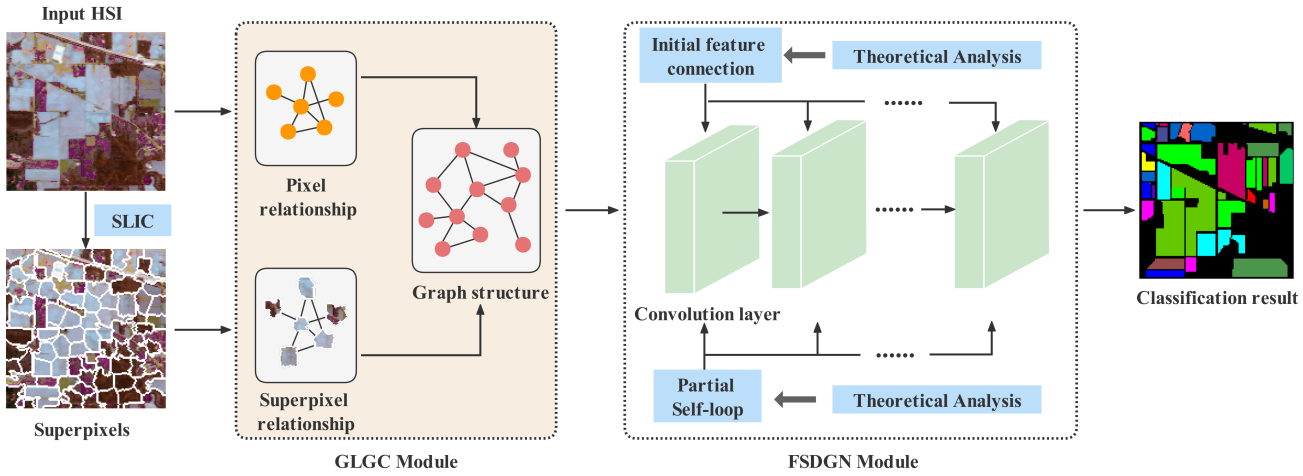

Fig. 1. Overview of the proposed FSDGN model. First, the superpixels are obtained by the SLIC function, and the relationships between pixels and the relationships between superpixels are calculated by the GLGC module, to obtain the topologically consistent graph structure. Subsequently, the node features are propagated through the graph by the first-order smoothing-based convolution layer of the FSDGN module, where a partial self-loop and an initial feature connection operation are added. After K layers, where K is the number of convolution layers, the final discriminative graph representation $\mathbf{H}^K$ is learned for classification.

A. Global and Local Topological Consistency-Based Graph Construction

Operating on graph-structured data, GCN gradually integrates the connectivity patterns and feature attributes [44]. In view of the grid structure of HSI data, the first step of applying GCN-based methods to HSI classification is certainly building a reliable graph structure for the original image.

Given the occurrence of the spectral variation in HSIs, the graph construction methods that rely solely on spatial–spectral similarity may be incapable of constructing sufficiently accurate adjacency relationships between nodes that are sufficiently accurate. Inspired by Ding et al. [39], we propose the GLGC module to generate a global and local consistent graph by fully using the topological relationships between pixels. Initially, the SLIC algorithm [45] is applied to segment HSIs into superpixels for enhanced local smoothness and a wider field of view. Let M denote the number of superpixel patches, and the spectral signature $\mathbf{P} \in \mathbb{R}^{M\times c} = [\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_M]^T$ along with the spatial information $\mathbf{Q} \in \mathbb{R}^{M\times 2} = [\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_M]^T$ of these M superpixels can be calculated by averaging the spectral feature and the spatial coordinates of the pixels inside each superpixel. Recall that we denote the input HSI by $\mathbf{X} \in \mathbb{R}^{N\times c}$, as well as its spatial coordinates notated by $\mathbf{T} \in \mathbb{R}^{N\times 2}$. For pixel i and pixel j, becoming closer in spectral measurement $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2$ and spatial distance $\|\boldsymbol{t}_i - \boldsymbol{t}_j\|_2^2$ should be of larger possibility to belong to the same class. We aim to build a matrix $\mathbf{Z} \in \mathbb{R}^{N\times N}$, which depicts the possibility of one pixel capturing another pixel, by solving the following mathematics model:

$$\min_{\mathbf{z}_i{}^{\mathrm{T}}\mathbf{1}=\mathbf{1}, 0\leqslant \mathbf{z}_i \leqslant 1} \sum_{i,j=1}^{N} \left(\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 + \varepsilon_p \|\boldsymbol{t}_i - \boldsymbol{t}_j\|_2^2\right)\mathbf{Z}_{ij} + \gamma_p \|\mathbf{Z}\|_F^2 \tag{6}$$

where $\|\mathbf{Z}\|_F^2$ is added to make sure the solution is nontrivial, and ε_p and γ_p are the regularization parameters. The resulted matrix $\mathbf{Z}$ concentrates on the relationships of input pixels, which builds a stronger connection between pixels with higher similarity.

However, the similarity matrix is not enough for pixels to capture all the intraclass pixels and build long-range connections. For example, two pixels x_i and x_j belonging to the same class may have a relatively low similarity in $\mathbf{Z}$ because they can be far apart in space. Considering this situation, we need another matrix $\mathbf{W}^p$ to describe the topological relationships between pixels and excavate the underlying adjacent relationships. The reason is that the topological structure of images remains unchanged, which is beneficial for mining potential neighbors belonging to the same class. The matrix $\mathbf{W}^p$ is naturally based on the rule that if pixel x_i and pixel x_j have high topological consistency, which means they are correlated by the topological structure of the graph, they tend to share some common neighbors. Based on this assumption, given the initial similarity matrix between pixels, the topological relationship can be found by searching the underlying useful neighbors. The mathematics model of $\mathbf{W}^p$ can then be defined as follows:

$$\min_{\mathbf{W}^p} \sum_{r}^{N} \left(\frac{1}{2} \sum_{i,j=1}^{N} \mathbf{Z}_{ij} \|\mathbf{W}_{ri}^p - \mathbf{W}_{rj}^p\|^2 + \eta_p \|\mathbf{W}_{ri}^p - \mathbf{I}_{ri}\|^2 \right) \tag{7}$$

where $\mathbf{W}_{ri}^p$ denotes the topological similarity between pixel $\boldsymbol{x}_r$ and pixel $\boldsymbol{x}_i$, and η_p is the regularization parameter. The first constrains that for every pixel $\boldsymbol{x}_r$, its topological relationship with pixel $\boldsymbol{x}_i$ and pixel $\boldsymbol{x}_j$ should be similar if $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ are assumed to be closely related by $\mathbf{Z}$, and the second term $\|\mathbf{W}_{ri}^p - \mathbf{I}_{ri}\|_2^2$ plays the same role as $\|\mathbf{Z}\|_F^2$ to avoid trivial solution.

As a result, for superpixels, the similarity matrix $\mathbf{G}$ and the topological relationship matrix $\mathbf{W}^s$ can be defined as follows:

$$\min_{\mathbf{g}_i{}^{\mathrm{T}}\mathbf{1}=1, 0\leqslant \mathbf{g}_i \leqslant 1} \sum_{i,j=1}^{M} \left(\|\boldsymbol{p}_i - \boldsymbol{p}_j\|_2^2 + \varepsilon_s \|\boldsymbol{q}_i - \boldsymbol{q}_j\|_2^2\right)\mathbf{G}_{ij} + \gamma_s \|\mathbf{G}\|_F^2 \tag{8}$$

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$$\min_{\mathbf{W}^s} \sum_{r=1}^{M} \left(\frac{1}{2} \sum_{i,j=1}^{M} \mathbf{G}_{ij} \|\mathbf{W}^s_{ri} - \mathbf{W}^s_{rj}\|^2 + \eta_s \|\mathbf{W}^s_{ri} - \mathbf{I}_{ri}\|^2 \right) \tag{9}$$

where $\boldsymbol{p}_i$ and $\boldsymbol{q}_i$ denote the average spectral signature and average spatial coordinate of the ith superpixel, respectively. The topological relationship between superpixels provides a broader view, which is conducive for pixels being far apart to be connected from a global view.

Subsequently, a probability matrix $\mathbf{S}$ is constructed to define the correlation between pixels and superpixels, or the possibility of a pixel being captured by a superpixel. $\mathbf{S}$ can be defined in the same pattern as $\mathbf{Z}$ and $\mathbf{G}$ by

$$\min_{\mathbf{s}_i{}^{\mathrm{T}}\mathbf{1}=\mathbf{1}, 0 \leqslant \mathbf{s}_i \leqslant 1} \sum_{i=1}^{N} \sum_{j=1}^{M} \left(\|\boldsymbol{x}_i - \boldsymbol{p}_j\|_2^2 + \varepsilon \|\boldsymbol{t}_i - \boldsymbol{q}_j\|_2^2 \right) \mathbf{S}_{ij} + \gamma \|\mathbf{S}\|_F^2 \tag{10}$$

where $\|\mathbf{S}\|_F^2$ is the regularization term for nontrivial solution, and ε and γ are the regulation parameters. We can solve problem (7) and (9) to obtain $\mathbf{W}^p$ and $\mathbf{W}^s$. Consequently, we define the adaptive topological attraction graph $\mathbf{B} \in \mathbb{R}^{N \times M}$, which is different from [39], by

$$\mathbf{B} = \pi \mathbf{S}\mathbf{W}^s + (1 - \pi)\mathbf{W}^p\mathbf{S}. \tag{11}$$

The two terms above can all be interpreted as the attraction of pixels to superpixels, while the first term shows that pixels topologically associated with pixel x_i contribute to which superpixel it belongs to, and the second term indicates that the possibility of being captured for pixel x_i can be passed through the flow of superpixel topological relationship, with a hyperparameter π balancing the effect of pixel topological relationship and superpixel topological relationship.

Finally, we define the ultimate normalized adjacent matrix for convolutional operations by

$$\widetilde{\mathbf{B}} = \hat{\mathbf{D}}^{-\frac{1}{2}} \left(\mathbf{B}\Delta\mathbf{B}^{\mathrm{T}} + \mu\mathbf{I} \right) \hat{\mathbf{D}}^{-\frac{1}{2}} \tag{12}$$

where μ is a predefined hyperparameter set to 0.1, and $\Delta_{ii} = \sum_{j=1}^{N} \mathbf{B}_{ji}$, $\hat{\mathbf{D}} = \mathrm{diag}(d_1, d_2, \ldots, d_n)$ where $d_i = \sum_j (\mathbf{B}\Delta\mathbf{B}^{\mathrm{T}} + \mu\mathbf{I})_{ij}$. By performing $\mathbf{B}\Delta\mathbf{B}^{\mathrm{T}}$, pixels spatially separated but possibly belonging to the same class are linked through the topological connectivity between pixels and superpixels, yielding our global and local topologically consistent graph. The whole process of graph construction in this study is summarized in Algorithm 1.

Algorithm 1 Graph Construction Process of FSDGN

Input: the spectral signature $\mathbf{X} \in \mathbb{R}^{N \times c} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]^T$ of pixels, the spatial coordinates $\mathbf{T} \in \mathbb{R}^{N \times 2} = [\boldsymbol{t}_1, \boldsymbol{t}_2, \ldots, \boldsymbol{t}_N]^T$ of pixels, the spectral feature $\mathbf{P} \in \mathbb{R}^{M \times c} = [\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_M]^T$ of superpixels, the spatial coordinates $\mathbf{Q} \in \mathbb{R}^{M \times 2} = [\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_M]^T$ of superpixels.

Output: the local and global topological consistent graph $\widetilde{\mathbf{B}}$

1: Calculate the topological relationship between pixels $\mathbf{W}^p$ by (7).
2: Calculate the topological relationship between superpixels $\mathbf{W}^s$ by (9).
3: Calculate the topological attraction graph between pixels and superpixels $\mathbf{S}$ by (10)
4: Calculate the normalized adjacent matrix $\widetilde{\mathbf{B}}$ by (11) and (12)
5: **return** $\widetilde{\mathbf{B}}$

B. First-Order Smoothing-Based Deep Graph Network

As a GCN model goes deeper, the representations of all the connected nodes in the graph tend to become entirely unrelated to the initial feature, and this is known as the oversmoothing problem. Therefore, it is reasonable to mix shallow representations that have not been oversmoothed with deep undistinguishable representations to obtain distinguishable feature expression. On one hand, the work of Xu et al. [46] has proved that integrating feature information from previous layers with graph convolution in deep layers improves the classification performance; on the other hand, adding feature information from all the previous layers consumes a substantial amount of computing in a relatively deep graph network, while introducing redundant or oversmoothed information that is detrimental to obtaining a discriminate graph representation. To address this problem, we propose the FSDGN. For each graph convolutional layer, we construct a residual connection from the last layer, an initial connection from the input layer, and intuitively add a skip connection from the first graph convolutional layer, which can be expressed as follows:

$$\mathbf{H}^l = \sigma\left(\mathrm{Aggregation}\left(\widetilde{\mathbf{B}}\mathbf{H}^{l-1}\Theta_l, \mathbf{H}^{l-1}, \mathbf{H}^0, \mathbf{H}^1 \right) \right) \tag{13}$$

where $\mathbf{H}^l$ denotes the graph representation of layer l, and $\Theta_l \in \mathbb{R}^{n \times n}$ is a learnable weight matrix. Here, $\mathbf{H}^0 = \Phi(\mathbf{X})$, and $\Phi \in \mathbb{R}^{c \times n}$ is a fully connected layer that linearly transforms $\mathbf{X}$ to a lower dimension before forward propagation. Note that in this study, the hidden units of all the graph convolutional layers are set to be n.

Since $\mathbf{H}^1 = \widetilde{\mathbf{B}}\mathbf{H}^0\Theta_1$, we replace $\mathbf{H}^1$ by $\widetilde{\mathbf{B}}\mathbf{H}^0\Theta_l$, where the weight matrix Θ_1 is replaced with Θ_l to reduce computation, and two parameters $\alpha, \beta \in (0, 1)$ are added to adjust the contribution of each part. Finally, the sum aggregation strategy is implemented. Formally, we define the lth layer of FSDGN as

$$\mathbf{H}^l = \sigma\left(\left((1 - \alpha)\widetilde{\mathbf{B}} \star \Theta_l + \alpha\mathbf{I} \right) \left((1 - \beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0 \right) \right) \tag{14}$$

where $\star$ denotes that rather than performing direct multiplication, $\widetilde{\mathbf{B}}$ is multiplied with the weight matrix Θ_l after being multiplied with the feature matrix first. By increasing the value of α, neighbor information aggregated through the adjacency matrix $\mathbf{B}$ is less involved in the representation of the present layer, while the residual information from the initial image and the first layer accounts for a larger proportion of the graph representation as the value of β becomes larger.

The addition of $\mathbf{H}^{l-1}$ in the proposed model ensures that the performance of the network at layer l is at least comparable to the performance of the previous layer, and the connection to $\mathbf{H}^0$ ensures that the final graph representation is bound to incorporate a part of the input data regardless of the

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number of layers stacked. By adding $\mathbf{H}^1$, we emphasize the importance of the information from the first-order neighborhood. As previously mentioned, because of the presence of the spectrum variation and spectral distortion, HSI typically possesses a high signal-to-noise ratio. Therefore, the initial features of HSI may be inaccurate sometimes due to the presence of noise. Differing from the original feature, the node representation in $\mathbf{H}^1$ can be viewed as local smoothing information obtained by first-order aggregation. On one hand, it has the effect of strengthening local features, so that these discriminative local features are retained in the node representation of deep layers, thus preventing oversmoothing. On the other hand, feature representations from $\mathbf{H}^1$ are more reliable than the original feature, since the act of aggregating information from the one-order neighborhood can serve as a kind of local smoothness, making the node representation approximate to the average feature within a class. Moreover, the credibility of first-order neighbors is guaranteed by our global and local topologically consistent graphs, which demonstrate the necessity of our graph construction method in turn. We also show the effect of $\mathbf{H}^1$ by experiments in Section V.

There are other smoothing-based approaches for HSI classification [47], [48], [49], [50]. Most of these methods operate directly on the spectral information and spatial information of HSIs, and then the more smoothed features are obtained for subsequent classification task. The smoothing operation in these methods is based on the transformation of feature information, especially transforming the spectral features. Comparing with these methods, the proposed FSDGN works in a different way. The local smoothing effect of FSDGN originates from aggregating local pixels' features based on the topological relationships between pixels, which are calculated by the spectral and spatial features. In other words, the smoothing operation does not rely on transforming the initial features, but the gathered information from topologically related pixels has a smoothing effect. As a result, our proposed FSDGN smoothes the input image in a different way from the aforementioned methods. The FSDGN can be extended using the image processed by these smoothing-based methods as the input graph. After that, the first-order smoothing-based layer can subsequently further smoothen the local features and extract high-order effective information.

Like GCN [43], after the graph convolutional operations, we eventually apply the cross-entropy loss to penalize the deviation of the network output from the labels of training data

$$\mathcal{L} = -\sum_{i\in\Upsilon}\sum_{j=1}^{C}\mathbf{X}_{ij}^{\text{out}}\ln\left(\mathbf{Y}_{ij}\right) \tag{15}$$

where Υ is the set of node indices that have labels, $\mathbf{X}^{\text{out}}$ is the network output, and $\mathbf{Y}$ is the label matrix. And the gradient descent algorithm is performed with the full training set in every iteration.

In general, the process of the proposed method is summarized in Algorithm 2.

Algorithm 2 Propagation Procedure of FSDGN

Input: The input graph feature $\mathbf{X}$, the adjacent matrix $\widetilde{\mathbf{B}}$.
Output: The network output $\mathbf{X}^{out}$

$\mathbf{H}^0 = \Phi(\mathbf{X})$
for $l = 1, 2, \ldots, K$ **do**
  $\mathbf{H}^l = ((1-\alpha)\widetilde{\mathbf{B}} \star \Theta_l + \alpha\mathbf{I})((1-\beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0)$
End for
$\mathbf{X}^{out} = softmax(\mathbf{H}^K)$
return $\mathbf{X}^{out}$

IV. Spectral Analysis of FSDGN

The oversmoothing problem, which can be interpreted as the indistinguishability of node representations, is the main difficulty of training a deep graph network. From the perspective of the spectral domain, we analyze why FSDGN can prevent the oversmoothing problem. We begin with the basic GCN.

Given the normalized graph Laplacian $\widetilde{\mathbf{L}}_{\text{sym}}$ of $\widetilde{\mathbf{B}}$, we use λ_i to denote the ith eigenvalue of $\widetilde{\mathbf{L}}_{\text{sym}}$. Then we can define the graph convolution on an input signal $x \in \mathbb{R}^c$ by a spectral filter g_ξ (parameterized by the eigenvalues of $\widetilde{\mathbf{L}}_{\text{sym}}$), which can be approximated in the Kth order polynomials form of normalized Laplacians by

$$g_\xi * x \approx \sum_{k=0}^{K}\xi_k\widetilde{\mathbf{L}}_{\text{sym}}^k x \tag{16}$$

where $\xi \in \mathbb{R}^{K+1}$ is a coefficient vector. As $\widetilde{\mathbf{L}}_{\text{sym}} = \mathbf{I} - \widetilde{\mathbf{B}}$, we can replace $\widetilde{\mathbf{L}}_{\text{sym}}$ with $\widetilde{\mathbf{B}}$ to obtain that

$$g_\omega * x = \sum_{k=0}^{K}\omega_k\widetilde{\mathbf{B}}^k x \tag{17}$$

where $\omega \in \mathbb{R}^{K+1}$ is a coefficient vector, and $\omega_k = \sum_{i=k}^{K}\xi_k\mathrm{C}_i^k(-1)^k$.

By stacking K layers, we can express the representation of GCN by

$$\mathbf{H}^K = \widetilde{\mathbf{B}}^K\mathbf{X}\Theta \tag{18}$$

where $\Theta = \Theta_1\Theta_2, \ldots, \Theta_K$. The activation function ReLU is removed to better discuss the filter of GCN. Compared with (17), a K-layer GCN only includes information from the term of order K in $\sum_{k=0}^{K}\omega_k\widetilde{\mathbf{B}}^k x$, and therefore its expression ability is largely impaired. On the contrary, a K-layer FSDGN that simulates (17) with a learnable ω_k is capable of including information from the input graph to the Kth order neighborhood, and this is the key to why it remains discriminative. Other deep graph network models that can be expressed in this way include approximate personalized propagation of neural predictions (APPNP) [51], GCNII [37], and DropEdge [34].

According to [36], Liu et al. have proved that the oversmoothing problem is inevitable in very deep models, which means we cannot completely solve the problem, but can only slow its convergence speed. We illustrate how FSDGN reduces the convergence rate in two ways: *partial self-loop* and *initial feature connection*.

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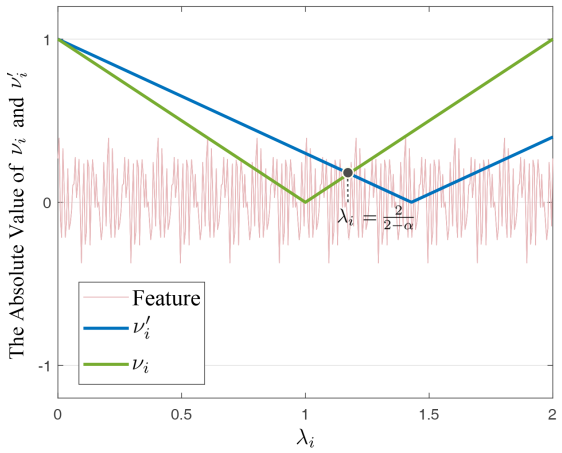

Fig. 2. Absolute value of ν_i and ν_i' with the change in λ_i.

A. Partial Self-Loop

To start with, we focus on how the oversmoothing problem occurs. With the eigendecomposition $\widetilde{\mathbf{B}} = \mathbf{U}'\Delta\mathbf{U}'^T$, the kth power of $\widetilde{\mathbf{B}}$ can be computed by

$$\widetilde{\mathbf{B}}^k = \mathbf{U}'\Delta^k\mathbf{U}'^T = \sum_{i=1}^{N} \nu_i^k \boldsymbol{\rho}_i \boldsymbol{\rho}_i^T \tag{19}$$

where Δ is a diagonal matrix of its eigenvalues, ν_i is the ith eigenvalue in Δ, and $\boldsymbol{\rho}_i$ is the corresponding normalized right eigenvector. The work of [52] proved that $\lambda_i \in [0, 2)$. Since $\widetilde{\mathbf{B}} = \widetilde{\mathbf{L}}_{\text{sym}} - \mathbf{I}_n$, we can deduce that the relationship between ν_i and λ_i is $\nu_i = 1 - \lambda_i$, $\nu_i \in (-1, 1]$. When the number of layers goes to infinity, the terms including $|\nu_i| < 1$ are reduced to 0, and only the term with an eigenvalue 1 is retained, i.e., $\lim_{k\to\infty} \widetilde{\mathbf{B}}^k = \boldsymbol{\rho} \cdot \boldsymbol{\rho}^T$, where $\boldsymbol{\rho}$ is the normalized right eigenvector of eigenvalue 1. According to [36], the right eigenvector of eigenvalue 1 is $\widetilde{\mathbf{D}}^{(1/2)}\mathbf{e}^T \in \mathbb{R}^{N\times 1}$. As a result, we have

$$\lim_{k\to\infty} \widetilde{\mathbf{B}}^k = \Phi\left(\widetilde{\mathbf{D}}^{\frac{1}{2}}\mathbf{e}^T\right)\left(\Phi\left(\widetilde{\mathbf{D}}^{\frac{1}{2}}\mathbf{e}^T\right)\right)^T \tag{20}$$

where $\Phi(x) = (x/\|x\|)$ is the normalization function, and $\|x\|$ denotes the Euclidean norm of x. The result of $\Phi(\widetilde{\mathbf{D}}^{(1/2)}\mathbf{e}^T)(\Phi(\widetilde{\mathbf{D}}^{(1/2)}\mathbf{e}^T))^T$ is a matrix with each row is proportional to the degree of the corresponding node, showing that the output graph is completely independent of the input feature $\mathbf{X}$. That is to say, if K is large enough, all the graphs sharing a common graph structure become the same after K times graph convolution, despite their different features, and thus lead to oversmoothing. Based on the basis of this conclusion, looking back at (18) and (19), we can easily find that when the model is deep, preserving terms with an eigenvalue $|\nu_i| < 1$ is the key to alleviating the oversmoothing problem.

We then consider a weak version of FSDGN by simplifying $((1-\beta)\mathbf{H}_{k-1}+\beta\mathbf{H}_0)$ as $\mathbf{H}_{k-1}$. Note that the activation function is put aside here to better discuss the filter of FSDGN. Now we have

$$\mathbf{H}^k = \left((1-\alpha)\widetilde{\mathbf{B}} \star \Theta_k + \alpha\mathbf{I}\right)\mathbf{H}^{k-1}. \tag{21}$$

As Θ_k is multiplied by $\widetilde{\mathbf{B}}$ after $\widetilde{\mathbf{B}}$ is multiplied by $\mathbf{H}^{k-1}$, (21) is expanded to

$$\mathbf{H}^k = \left((1-\alpha)\widetilde{\mathbf{B}} + \alpha\mathbf{I}\right)\mathbf{H}^{k-1}\Theta_k. \tag{22}$$

Let $\widetilde{\mathbf{B}}' = (1-\alpha)\widetilde{\mathbf{B}} + \alpha\mathbf{I}$, where $\alpha \in (0, 1)$. Equation (22) can be regarded as a graph convolution operation with a filter $\widetilde{\mathbf{B}}'$. We can consider $\widetilde{\mathbf{B}}'$ as introducing a partial self-loop in $\widetilde{\mathbf{B}}$. Then we show how the filter of $\widetilde{\mathbf{B}}'$ slows down the convergence with Theorem 1.

Theorem 1: For any eigenvalue of $\widetilde{\mathbf{B}}'$, $\widetilde{\mathbf{B}}$, and the normalized graph Laplacian matrix $\widetilde{\mathbf{L}}_{\text{sym}}$, we have

$$0 \leqslant |\nu_i| < |\nu_i'| \leqslant 1 \quad \left(\lambda_i < \frac{2}{2-\alpha}\right) \tag{23}$$

where ν_i is the ith eigenvalue of $\widetilde{\mathbf{B}}$, ν_i' is the ith eigenvalue of $\widetilde{\mathbf{B}}'$, and λ_i is the ith eigenvalue of $\widetilde{\mathbf{L}}_{\text{sym}}$.

Proof. With $\nu_i = 1-\lambda_i$, $\nu_i \in (-1, 1]$, we can also conclude that the relationship between ν_i' and λ_i is $\nu_i' = (1-\alpha)\nu_i + \alpha = 1 - (1-\alpha)\lambda_i$. As $\alpha \in (0, 1)$, the value of ν_i' must be larger than ν_i, but less than 1. For $\lambda_i \leqslant 1$, $\nu_i' > \nu_i \geqslant 0$. Nevertheless, for $\lambda_i > 1$, $\nu_i < 0$ and at the same time $\nu_i < \nu_i'$. We can set up the following equation to ensure $|\nu_i| < |\nu_i'|$ by limiting the value of λ_i:

$$1 - (1-\alpha)\lambda_i > -(1-\lambda_i) \tag{24}$$

of which the solution is $\lambda_i < (2/2-\alpha)$. The effect of partial self-loop is visualized in Fig. 2.

As proved by Theorem 1, the FSDGN model achieves the effect of preserving terms with an eigenvalue of $|\nu_i'| < 1$ by increasing the value of $|\nu_i'^k|$; consequently, the convergence of terms that satisfy the condition in (23) converges to 0 at a slower rate. From a different perspective, Wu et al. [52] state that the filter of GCN functions as a low-pass-type filter as the model goes deeper so that the information carried by eigenvalues greater than 1 is weakened in deep layers, but it also results in a loss of information. FSDGN also reduces this loss by compressing the information loss interval from (1,2) to $[(2/2-\alpha), 2]$ for λ_i, allowing more information to pass. Moreover, the FSDGN differs from (21) in that it separates the processes of residual information connection and feature transformation, thereby preventing the unreasonable intertwining of residual information with the current layer parameters. However, it is clear that this solution cannot solve the oversmoothing problem, especially when the model is extremely deep, as $|\nu_i'^k|$ will eventually be almost equal to 0 when k approaches infinity. This also explains why the residual GCN [43] has an apparent drop in accuracy when layers are particularly deep.

B. Initial Feature Connection

Similarly, we weaken the FSDGN by letting $((1-\alpha)\widetilde{\mathbf{B}} \star \Theta_l + \alpha\mathbf{I})$ to be $\widetilde{\mathbf{B}} \star \Theta_l$, to focus on how the term of $((1-\beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0)$ affects the deep graph convolution. The term of $\beta\mathbf{H}^0$ here ensures that part of the initial feature is included in the input feature of layer l, no matter how many layers are stacked, and this operation is named "initial feature connection." This simplified FSDGN can be rewritten as

$$\mathbf{H}^l = \widetilde{\mathbf{B}}\left((1-\beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0\right)\Theta_l. \tag{25}$$

Assuming that the input feature is nonnegative, the K-order graph representation of (25) can be defined as

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$$\mathbf{H}^K = \left(\sum_{k=1}^{K} \epsilon_k \widetilde{\mathbf{B}}^k\right)\mathbf{H}^0\Theta \tag{26}$$

where $\epsilon_k = \beta(1-\beta)^{k-1}$ for $k \neq 1$ or K, and $\epsilon_1 = \beta, \epsilon_K = (1-\beta)^{K-1}$. We can further replace $\widetilde{\mathbf{B}}^k$ via (19), and $\mathbf{H}^K$ can be represented as

$$\begin{aligned}\mathbf{H}^K &= \left(\sum_{k=1}^{K} \epsilon_k \sum_{i=1}^{N} \nu_i^k \boldsymbol{\rho}_i \boldsymbol{\rho}_i{}^T\right)\mathbf{H}^0\Theta \\ &= \left(\sum_{i=1}^{N}\left(\sum_{k=1}^{K} \epsilon_k \nu_i^k\right)\boldsymbol{\rho}_i \boldsymbol{\rho}_i{}^T\right)\mathbf{H}^0\Theta. \end{aligned} \tag{27}$$

In a Kth order GCN, for those $|\nu_i| < 1$, the coefficient of its eigenvector is ν_i^k, whereas in (27), the corresponding coefficient is $\sum_{k=1}^{K} \epsilon_k \nu_i^k$. It is obvious that there is always a term of $\epsilon_1 \nu_i$, i.e., $\beta\nu_i$ in $\sum_{k=1}^{K} \epsilon_k \nu_i^k$. When k is very large, the absolute value of $\epsilon_1 \nu_i$ is easily greater than ν_i^k, even with a small β, and thus the value of $\sum_{k=1}^{K} \epsilon_k \nu_i^k$ is greater than ν_i^k. So, when the initial feature connection is used, the information carried by these eigenvectors is amplified, and to some extent, the disappearance of such information in deep layers can be prevented.

Fig. 3. Indian Pines dataset. (a) False-color image. (b) Ground-truth image.

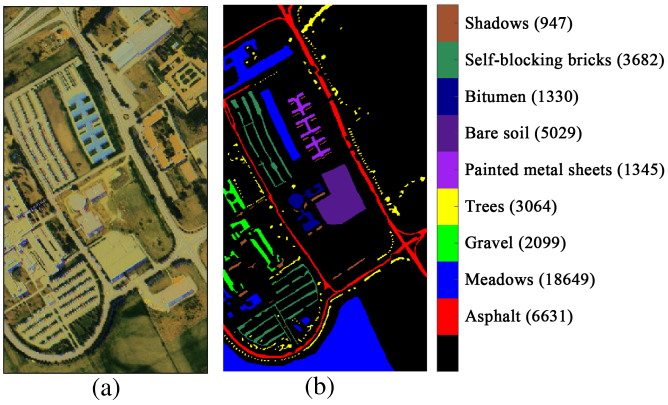

Fig. 4. PaviaU dataset. (a) False-color image. (b) Ground-truth image.

V. Experimental Results and Analysis

In this section, experiments were performed to investigate and validate the performance of FSDGN on three HSI datasets, and various state-of-the-art methods were compared with the proposed FSDGN. For comparison, we chose one traditional classification method, joint collaborative representation and SVM with decision fusion (JSDF) [53], two CNN-based methods, deep pyramidal residual network (DPRN) [21] and a CNN with pixel-pair features (CNN-PPF) [54], and several GCN-based methods, including GCN [43], spectral–spatial GCN (S^2GCN) [24], multiscale dynamic GCN (MDGCN) [27], context-aware dynamic GCN (CAD-GCN) [55], globally consistent GCN (GCGCN) [39], and novel deep network with adaptive graph structure integration (DNSGSI) [38]. The overall accuracy (OA), the average accuracy (AA), and the kappa coefficient (KC) were adopted as the evaluation metrics. In addition, we conducted experiments of FSDGN using different sample sizes to show its effectiveness. The performance of FSDGN at various depths has also been tested. In addition, the ablation tests were performed to demonstrate the effect of the first-order smoothing information in FSDGN, and we analyzed how the parameters in FSDGN affected the classification accuracy.

A. Dataset Description

There are three datasets used in our experiments, and below are their introductions.

1) *Indian Pines:* Indian Pines was the first test data used for HSI classification. In 1992, the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) imaged a patch of Indian pines in IN, USA. AVIRIS imagined the land features in a range of 0.4–2.5 μm in 220 consecutive wavebands, with a spatial resolution of 20 m. The imaging results were cropped to a size of 145 × 145 and were labeled for HSI classification. Since there are 20 bands that cannot be reflected by water, we generally use the remaining 200 bands after removing these 20 bands for experiments. This image contains pixels from 16 different classes, and Fig. 3 shows the false-color image and the ground-truth image of Indian Pines.
2) *Pavia University (PaviaU)*: PaviaU is the HS data of Pavia city, which was collected by the Reflective Optics Spectrographic Imaging System (ROSIS-03) of German in 2003. The spectral imager continuously imaged 115 bands in the wavelength range of 0.43–0.86 μm with a spatial resolution of 1.3 m. Among them, 12 bands were removed due to the influence of noise, so the image composed of the remaining 103 spectral bands is generally used. The PaviaU dataset includes nine classes with a total size of 610 × 340. Fig. 4 shows the corresponding false-color image and the ground-truth image. Fig. 4 shows the corresponding false-color image and the ground-truth image.
3) *Houston University (Houston)*: The Houston dataset was provided by the National Center for Airborne Laser Mapping (NCALM), a National Science Foundation (NSF) funded project at the University of Houston, USA. The dataset was acquired over the University of Houston campus and the neighboring urban area and was originally used for scientific purposes in the 2013 IEEE GRSS Data Fusion Competition. The size of the Houston dataset is 349C-1905, and its spatial resolution is 2.5 m and contains 144 spectral bands. In addition, it has 15 classes including healthy grass, artificial turf, etc.

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TABLE I

NUMBERS OF TRAINING AND TESTING SAMPLES OF EACH CLASS ON THE INDIAN PINES, PAVIAU, AND HOUSTON DATASETS

Indian Pines				PaviaU				Houston			
class	name	train	test	class	name	train	test	class	name	train	test
1	Alfalfa	30	16	1	Asphalt	30	6601	1	Healthy grass	30	1211
2	Corn-notill	30	1398	2	Meadows	30	1861	2	Stressed grass	30	1224
3	Corn-mintill	30	800	3	Gravel	30	2069	3	Synthetic grass	30	667
4	Corn	30	207	4	Trees	30	3034	4	Tree	30	1214
5	Grass-pasture	30	453	5	Painted metal sheets	30	1345	5	Soil	30	1212
6	Grass-trees	30	700	6	Bare Soil	30	4999	6	Water	30	295
7	Grass-pasture-mowed	15	13	7	Bitumen	30	1300	7	Residential	30	1238
8	Hay-windrowed	30	448	8	Self-Blocking Bricks	30	3652	8	Commercial	30	1214
9	Oats	15	5	9	Shadows	30	917	9	Road	30	1222
10	Soybean-notill	30	942		Total	270	42506	10	Highway	30	1197
11	Soybean-mintill	30	2425					11	Railway	30	1205
12	Soybean-clean	30	563					12	Parking lot 1	30	1203
13	Wheat	30	175					13	Parking lot 2	30	439
14	Woods	30	1235					14	Tennis court	30	398
15	Buildings-Grass-Trees-Drives	30	356					15	Running track	30	630
16	Stone-Steel-Towers	30	63						Total	450	14579
	Total	450	9799								

TABLE II

PER-CLASS ACCURACY, OA, AA (%), AND KC OF DIFFERENT CLASSIFICATION METHODS ON THE INDIAN PINES DATASET

class	JSDF [53]	DPRN [21]	CNN-PPF [54]	GCN [43]	S^2GCN [24]	MDGCN [27]	CADGCN [55]	GCGCN [39]	DNAGSI [38]	FSDGN
1	**100(0)**	**100(0)**	95.00(2.64)	95.00(2.80)	**100(0)**	**100(0)**	99.46(1.54)	**100(0)**	98.25(0.43)	96.25(1.68)
2	90.75(3.19)	74.6(8.51)	73.53(5.61)	56.71(4.42)	84.43(2.50)	80.18(0.84)	88.08(4.57)	91.15(3.70)	92.29(3.27)	**96.78**(3.18)
3	77.84(3.81)	85.42(5.54)	81.34(3.76)	51.50(2.56)	82.87(5.53)	**98.26(0)**	95.62(1.89)	92.53(4.47)	96.16(1.52)	97.53(2.58)
4	99.86(0.33)	99.12(0.83)	91.84(3.53)	84.64(3.16)	93.08(1.95)	98.57(0)	97.85(2.75)	**99.95(0.15)**	97.25(0.87)	98.74(3.25)
5	87.20(2.73)	93.34(4.26)	93.69(0.84)	83.71(3.20)	97.13(1.34)	95.14(0.33)	93.79(3.71)	**97.04(1.87)**	96.68(0.94)	96.11(1.23)
6	98.54(0.28)	97.24(2.88)	97.46(1.01)	94.03(2.11)	97.29(1.27)	97.16(0.57)	96.41(1.78)	96.79(3.93)	97.94(1.45)	**98.86**(0.49)
7	**100(0)**	**100(0)**	75.38(8.73)	92.31(0.00)	92.31(0)	**100(0)**	97.95(2.84)	96.15(4.05)	**100(0)**	96.92(4.42)
8	99.80(0.31)	99.18(0.91)	98.01(0.69)	96.61(1.86)	99.03(0.93)	98.89(0)	99.81(0.25)	**100(0)**	98.86(0.78)	**100(0)**
9	**100(0)**	**100(0)**	**100(0)**	**100(0)**	**100(0)**	**100(0)**	**100(0)**	**100(0)**	**100(0)**	**100(0)**
10	89.99(4.24)	85.53(4.61)	82.30(1.55)	77.47(1.24)	93.77(3.72)	90.02(1.02)	89.34(4.03)	91.51(4.57)	94.88(2.71)	**96.09**(4.32)
11	76.75(5.12)	75.34(4.17)	62.64(3.32)	56.56(1.53)	84.98(2.82)	93.35(1.47)	93.09(3.14)	95.38(2.74)	96.33(4.53)	**98.75(1.33)**
12	87.10(2.82)	79.04(4.96)	88.92(2.50)	58.29(6.58)	80.05(5.17)	93.05(2.30)	93.41(2.54)	**95.81(2.84)**	93.52(3.62)	95.13(3.79)
13	99.89(0.36)	99.91(0.23)	98.80(0.57)	**100(0)**	99.43(0)	**100(0)**	99.79(0.29)	99.31(0.53)	97.56(0.26)	99.20(0.30)
14	97.21(2.78)	92.92(3.55)	86.49(2.23)	80.03(3.93)	96.73(0.92)	99.72(0.05)	99.36(0.54)	99.26(1.37)	99.76(1.21)	**100(0)**
15	99.58(0.68)	96.07(3.04)	86.71(4.36)	69.55(6.66)	86.80(3.42)	**99.72(0)**	99.20(0.87)	97.5(2.72)	97.15(3.70)	98.99(2.27)
16	100(0)	99.74(0.65)	92.70(3.45)	98.41(0)	**100(0)**	95.71(0)	98.91(1.01)	99.52(0.77)	98.92(0.47)	**100(0)**
OA(%)	88.34(1.39)	84.89(1.70)	80.09(1.56)	69.24(1.56)	89.49(1.08)	93.47(0.38)	94.13(0.78)	95.35(0.95)	96.04(1.27)	**98.14(0.64)**
AA(%)	92.46(0.75)	92.34(0.94)	88.57(0.77)	80.93(1.71)	92.99(1.04)	96.24(0.21)	96.38(0.35)	96.80(0.84)	97.22(1.03)	**98.41**(0.85)
$\kappa \times 100$	86.80(1.55)	82.86(1.91)	77.52(1.74)	65.27(1.80)	88.00(1.23)	92.55(0.43)	93.29(0.88)	94.67(1.09)	94.94(1.68)	**97.79**(0.74)
Time(s)	518	1794	1495	58	71	95	62	641	652	826

B. Experimental Setup

In our experiments, the proposed FSDGN algorithm was implemented using PyTorch and the Adam optimizer. In the experiments of comparing FSDGN with other methods, we used a unified sample size setting of 30 for those classes with more than 30 examples and 15 for classes with less than 30 examples, as shown in Table I. During the training phase, the remaining pixels in each class were treated as unlabeled examples and then used as test sets to evaluate the classification performance. The hidden units n were set to be 200, and the learning rate was 0.008. In this study, all the results were achieved by taking the average of ten repetitions of the corresponding experiment, to eliminate the influence of random sampling. In addition, unless otherwise specified, the number of layers for FSDGN is set to 15.

C. Experimental Results

We compared the aforementioned models with the proposed FSDGN on the three HSI datasets introduced above. Here, we present the experimental results with analysis.

1) *Results on Indian Pines:* Table II shows the quantitative performance on Indian Pines of different methods, with the highest value of each class (i.e., in each row) highlighted in bold, and the corresponding classification maps are displayed in Fig. 6. The proposed FSDGN outperforms all the previous methods in terms of OA, AA, and KC. This indicates that our FSDGN captures high-order feature information and effective graph representation, thus contributing to the abilities of the classifier. In addition, the CNN-based models outperform GCN, demonstrating the benefit

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TABLE III
PER-CLASS ACCURACY, OA, AA (%), AND KC OF DIFFERENT CLASSIFICATION METHODS ON THE PAVIAU DATASET

class	JSDF [53]	DPRN [21]	CNN-PPF [54]	GCN [43]	S^{2}GCN [24]	MDGCN [27]	CADGCN [55]	GCGCN [39]	DNAGSI [38]	FSDGN
1	82.40(4.07)	92.83(1.29)	95.73(0.80)	69.78(4.71)	92.87(3.79)	93.55(0.37)	83.85(4.80)	94.59(2.57)	93.94(2.78)	**97.76**(3.01)
2	90.76(3.74)	95.16(2.29)	84.01(1.99)	54.10(10.54)	87.06(4.47)	99.25(0.23)	95.56(1.97)	98.11(3.49)	97.73(1.48)	**99.76(0.22)**
3	86.71(4.14)	87.36(6.47)	86.45(1.94)	69.69(4.48)	87.97(4.77)	92.03(0.24)	95.50(2.41)	99.35(1.76)	94.64(1.35)	**99.83**(0.41)
4	92.88(2.16)	94.12(3.66)	91.70(2.06)	91.23(7.02)	90.85(0.94)	83.78(1.55)	81.90(3.51)	96.11(1.96)	97.05(1.45)	**97.07**(2.71)
5	**100(0)**	99.70(0.40)	99.93(0.04)	98.74(0.11)	**100(0)**	99.47(0.09)	98.91(1.06)	99.77(0.09)	99.70(0.58)	99.82(0.38)
6	94.30(4.55)	82.63(26.14)	93.57(1.28)	65.34(10.53)	88.69(2.64)	95.26(0.50)	97.81(2.89)	**99.66(0.23)**	99.28(1.98)	99.55(0.10)
7	96.62(1.37)	87.48(6.96)	93.53(0.72)	86.64(4.68)	98.88(1.08)	98.92(1.04)	96.79(2.42)	99.53(0.91)	**99.92(0.31)**	99.75(0.64)
8	94.69(3.74)	93.44(1.13)	83.83(1.60)	72.26(2.63)	89.97(3.28)	94.99(1.33)	95.35(2.76)	97.84(1.50)	97.71(2.57)	**98.09**(2.04)
9	99.56(0.36)	97.99(1.62)	99.47(0.34)	**99.93(0.06)**	98.89(0.53)	81.03(0.49)	84.80(5.33)	96.80(1.07)	95.53(1.88)	97.53(3.20)
OA(%)	90.82(1.30)	92.69(2.36)	88.72(0.95)	66.19(3.43)	89.74(1.70)	95.68(0.22)	92.91(1.01)	97.71(1.55)	98.18(1.69)	**99.05**(0.49)
AA(%)	93.10(0.65)	92.30(3.24)	92.02(0.37)	78.63(1.23)	92.80(0.47)	93.15(0.28)	92.27(0.94)	97.98(0.41)	97.27(0.77)	**98.88**(0.48)
$\kappa \times 100$	88.02(1.62)	90.25(3.35)	85.43(1.18)	58.39(3.28)	86.65(2.06)	94.25(0.29)	90.69(1.29)	96.98(2.01)	97.59(1.57)	**98.73**(0.65)
Time(s)	1123	3669	1545	1783	1803	244	73	1653	772	1939

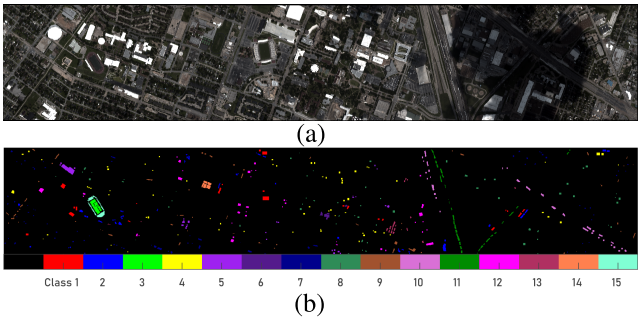

Fig. 5. Houston dataset. (a) False-color image. (b) Ground-truth image.

of involving spatial information in HSI classification. Moreover, the improved classification results achieved by the GCN-based models demonstrate the potential of the information aggregation pattern based on the graph structure in learning spatial–spectral feature representations. Among them, GCGCN is superior than other GCN-based methods, confirming that considering topological relationships and constructing global connections improves the quality of the graph structure. The proposed FSDGN outperforms GCGCN because it constructs long-range relationships between node samples through not only a topologically consistent graph but also a deep network structure, and FSDGN simultaneously learns that high-order discriminative features in the deep layer are learned.

2) *Results on PaviaU:* Table III summarizes the classification results of different models on the PaviaU dataset, and Fig. 7 visualizes the classification of different methods. Similarly, with notable increases in OA, AA, and KC, the proposed FSDGN outperforms previous methods. Compared with other methods, FSDGN reaches particularly high accuracy in most classes, such as asphalt, meadows, and gravel. In addition, in contrast to the circumstances on Indian Pines, the MDGCN performs better than CAD-GCN on this dataset. A possible explanation for this is that multiscale information plays an important role here.

3) *Results on Houston:* The experimental results on Houston are shown in Table IV, with the related classification maps shown in Fig. 8. Still, the proposed FSDGN is superior to the existing methods by a significant margin. The benefits of high-order advanced deep feature information extracted by FSDGN are further demonstrated.

4) *Computational Complexity and Running Time:* The running time of different models is shown in the bottom line in Tables II–IV. The computational complexity of Algorithm 1 to obtain the topologically ocnsistent graph is $O(N^2)$, and the computational complexity of Algorithm 2 is $O(K|\hat{N}|n^2)$, where K is the number of layers, and $|\hat{N}|$ represents the number of nonzero graph edges. As a result, the total computational complexity of the proposed FSDGN is $O(N^2 + K|\hat{N}|n^2)$. The primary computational complexity of FSDGN is from the construction of the topologically consistent graph. In addition, since FSDGN is a deep model, it is no doubt that more layers will increase the running time because of the convolution operation and backpropagation. However, unlike the MDGCN, once the topologically consistent graph has been constructed, it does not need to be dynamicly recalculated because it is adaptive to each layer, which saves time in each convolution layer. In addition, the experiments of FSDGN are performed with 15 layers. Due to the convolution operation and backpropagation, adding more layers will inevitably increase the time. Nevertheless, the total running time of FSDGN is neither the fastest nor the lowest. Moreover, the performance of FSDGN is the best compared with other methods described above. The time consumption of FSDGN is acceptable given the high accuracy it achieves.

D. Impact of the Number of Training Samples

To further evaluate the proposed FSDGN, we tested the impact on the overall classification accuracy with training samples of different numbers using the previously mentioned three

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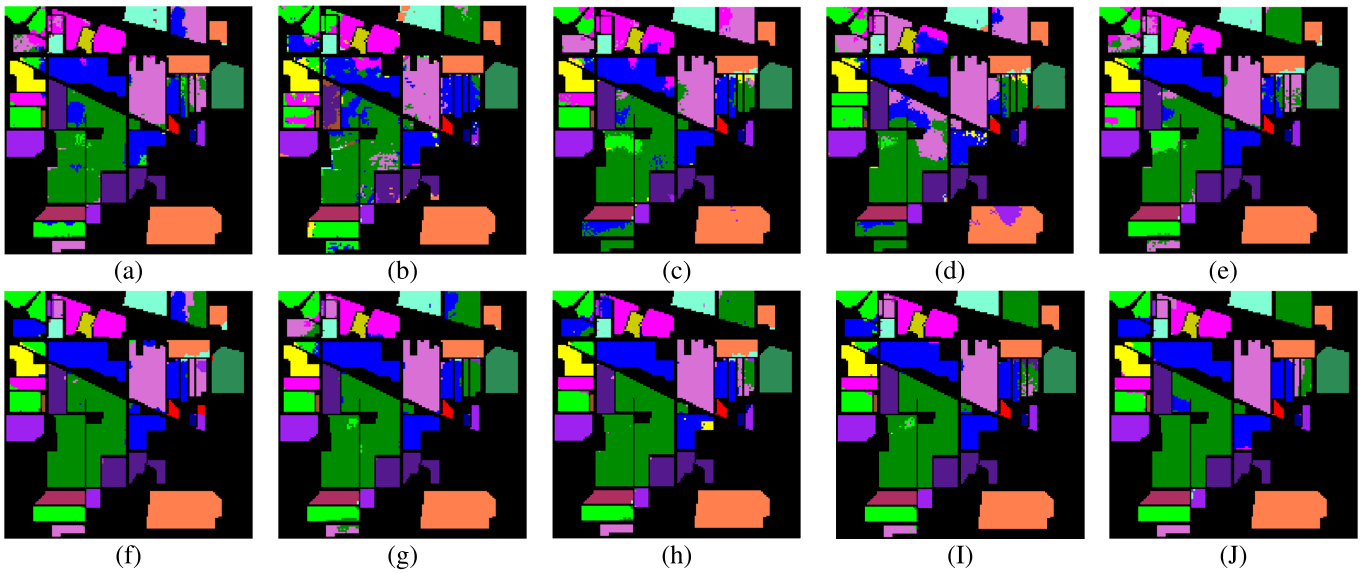

Fig. 6. Classification result images on the Indian Pines dataset. (a) JSDF. (b) DPRN. (c) CNN-PPF. (d) GCN. (e) S^2GCN. (f) MDGCN. (g) CAD-GCN. (h) GCGCN. (i) DNAGSI. (j) FSDGN.

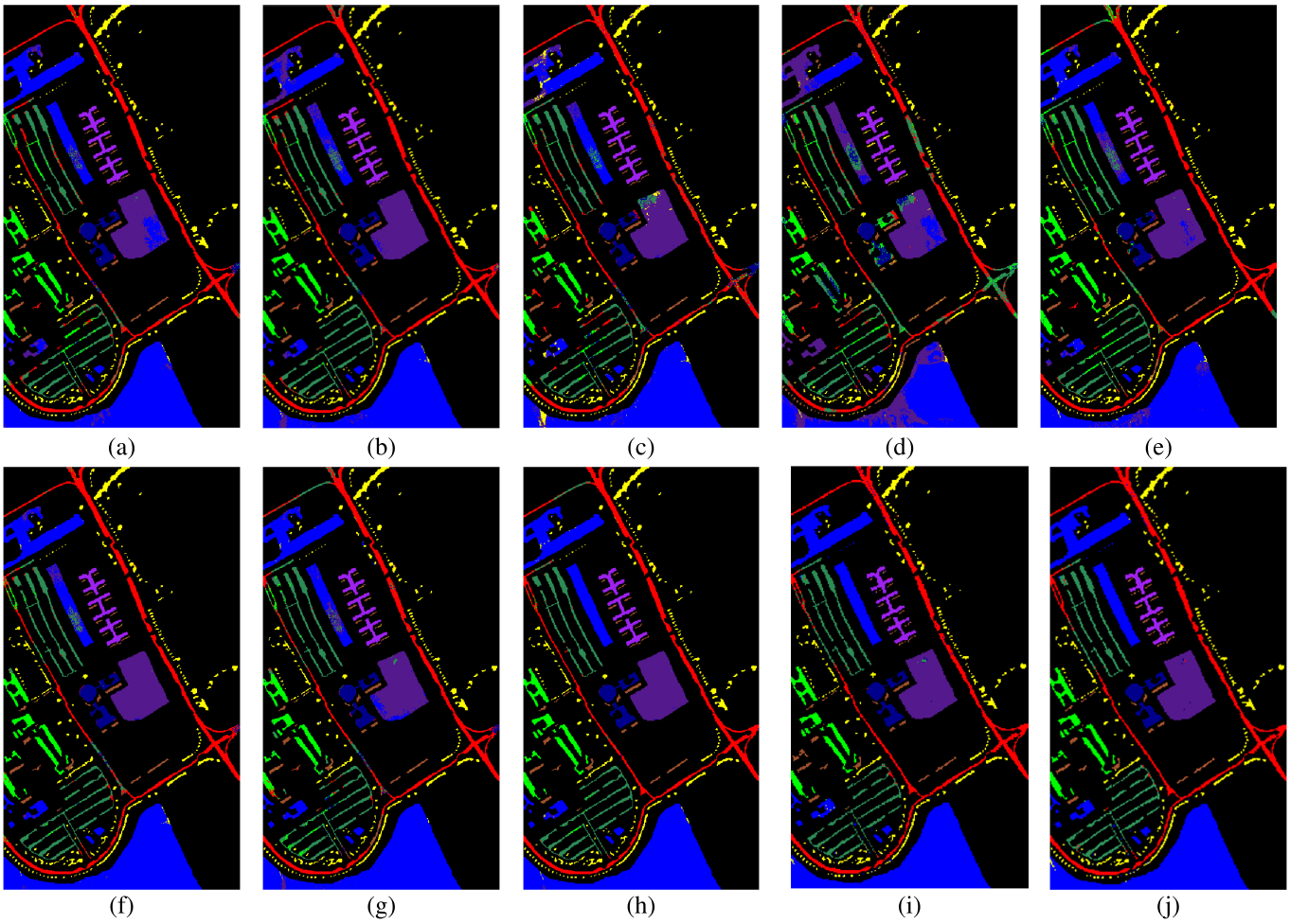

Fig. 7. Classification result images on the PaviaU dataset. (a) JSDF. (b) DPRN. (c) CNN-PPF. (d) GCN. (e) S^2GCN. (f) MDGCN. (g) CAD-GCN. (h) GCGCN. (i) DNAGSI. (j) FSDGN.

datasets, and the corresponding results are shown in Fig. 9. The number of training samples varies from 5 to 30, where an increase of five training samples is gradually added each time. From the chart, we can find that in all the cases, even when the training set is rather small, FSDGN always get higher accuracy than other methods. It also can be observed from Fig. 9 that when FSDGN is confronted with the fluctuation of the sample count, it shows stronger stability among all the models.

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TABLE IV
PER-CLASS ACCURACY, OA, AA (%), AND KC OF DIFFERENT CLASSIFICATION METHODS ON THE HOUSTON DATASET

class	JSDF [53]	DPRN [21]	CNN-PPF [54]	GCN [43]	S^{2}GCN [24]	MDGCN [27]	CADGCN [55]	GCGCN [39]	DNAGSI [38]	FSDGN
1	97.41(1.21)	97.53(1.45)	**98.62(0.71)**	88.16(1.90)	96.30(3.07)	93.42(4.25)	94.45(3.49)	92.30(2.75)	94.19(1.29)	94.10(3.10)
2	**99.48(0.25)**	97.83(1.40)	98.15(0.53)	97.20(0.48)	98.57(1.47)	93.67(3.60)	96.43(2.83)	96.08(2.70)	89.30(2.69)	97.08(1.47)
3	99.88(0.22)	**99.91(0.13)**	99.01(0.33)	97.91(0.13)	98.88(0.43)	98.12(1.09)	95.17(4.11)	99.40(0.52)	**100(0)**	99.86(0.39)
4	98.22(2.80)	94.93(1.46)	93.21(0.48)	96.55(0.41)	97.68(2.89)	95.58(1.85)	94.82(2.38)	97.47(1.79)	99.59(3.88)	**99.05**(2.64)
5	**100(0)**	99.40(0.96)	99.13(0.73)	89.79(0.71)	97.66(1.12)	99.00(1.30)	98.91(1.51)	97.91(1.23)	99.92(0.75)	99.28(1.65)
6	99.32(1.09)	95.19(3.50)	91.26(5.25)	98.21(1.15)	96.84(1.17)	93.28(6.08)	97.48(3.48)	98.51(1.10)	98.30(0.15)	**100(0)**
7	91.93(4.91)	88.56(1.89)	81.54(5.84)	73.67(1.94)	83.48(5.89)	87.68(4.41)	91.58(3.16)	**95.92**(3.42)	94.10(1.54)	93.93(2.98)
8	68.82(6.16)	79.64(4.72)	68.15(3.97)	65.71(4.64)	76.15(4.37)	80.45(6.12)	74.63(4.82)	81.37(4.79)	**87.07**(2.29)	84.81(4.83)
9	69.47(8.56)	81.51(4.93)	77.17(1.65)	70.27(3.03)	82.17(1.78)	89.64(2.26)	86.75(3.58)	**100(0)**	79.30(1.69)	85.73(1.41)
10	85.63(9.32)	93.40(2.91)	92.12(1.76)	74.71(2.32)	86.85(8.32)	90.06(6.41)	94.23(3.34)	**98.40**(1.25)	95.65(1.07)	98.39(1.60)
11	94.51(3.82)	89.86(4.59)	81.05(3.31)	75.36(2.37)	88.57(5.06)	86.73(3.22)	94.65(2.73)	98.38(1.32)	**99.75(0.56)**	**98.44**(2.44)
12	84.33(5.33)	92.58(4.40)	78.10(5.07)	79.29(4.80)	78.64(4.79)	89.44(5.69)	89.55(1.93)	89.93(5.38)	88.78(0.64)	**89.94**(5.27)
13	**98.10(1.28)**	96.95(1.57)	72.55(4.36)	79.29(4.80)	78.64(4.79)	89.44(5.69)	89.55(1.93)	75.17(3.35)	78.80(0.15)	74.89(2.07)
14	**100(0)**	**100(0)**	99.85(0.16)	86.03(3.31)	99.45(0.44)	99.43(0.97)	**100(0)**	**100(0)**	**100(0)**	**100(0)**
15	**99.86**(0.36)	99.84(0.23)	98.60(0.34)	95.29(1.67)	98.03(1.07)	96.27(1.72)	98.02(1.42)	99.45(0.31)	99.84(0.73)	99.81(0.37)
OA(%)	90.51(0.95)	92.73(0.65)	87.54(1.03)	80.35(0.61)	89.31(1.00)	91.40(0.92)	92.51(0.73)	93.85(0.40)	93.40(0.52)	**94.11**(0.59)
AA(%)	92.46(0.75)	93.64(0.61)	88.57(0.77)	80.02(0.46)	90.33(1.06)	92.37(0.89)	93.57(0.60)	93.91(0.40)	93.75(0.43)	**94.33**(0.58)
$\kappa \times 100$	92.14(0.70)	86.53(1.12)	77.52(1.74)	78.72(0.66)	88.44(1.08)	90.70(1.00)	91.89(0.78)	93.35(0.44)	92.87(0.24)	**93.74**(0.64)
Time(s)	621	1637	1223	901	971	360	159	835	648	1049

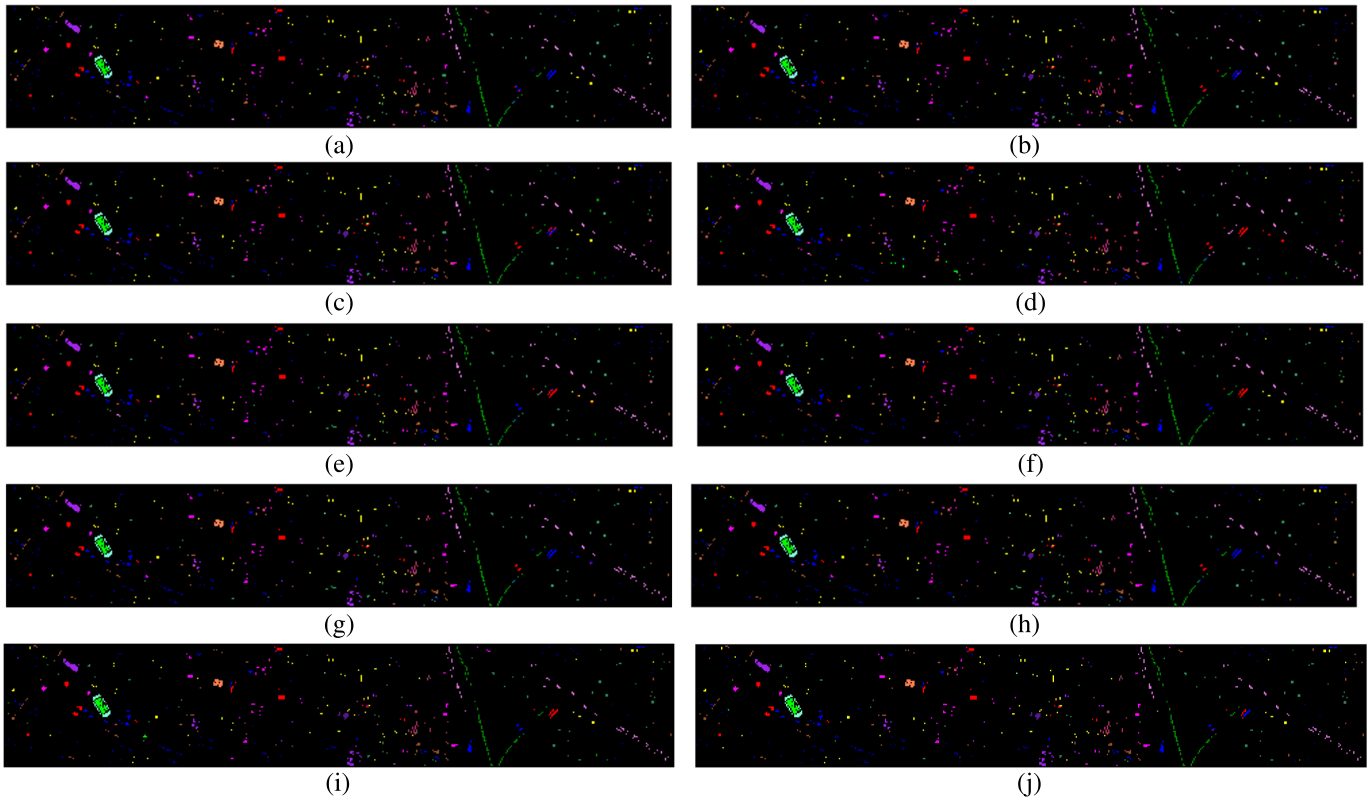

Fig. 8. Classification result images on the Houston dataset. (a) JSDF. (b) DPRN. (c) CNN-PPF. (d) GCN. (e) S^{2}GCN. (f) MDGCN. (g) CAD-GCN. (h) GCGCN. (i) DNAGSI. (j) FSDGN.

E. Ablation Studies

In Section III, we proposed that adding information from first-order neighbor aggregation, in other words, the graph representation of the first layer, can be helpful in HSI classification. Here, we present some experiments to test its contribution. We designed several variants of FSDGN and carried experiments. These variants include FSDGN$_{\text{res}}$, FSDGN$_{\text{h}_0}$, FSGDN$_{\text{h}_2}$, and FSGDN$_*$. FSDGN$_{\text{res}}$ denotes a normal GCN model with residual connection, and FSDGN$_{\text{h}_0}$ is a variant of FSDGN with the absence of $\mathbf{H}_1$ in each convolution layer. FSGDN$_{\text{h}_2}$ is an FSDGN-based model that extra information from $\mathbf{H}_2$ is also added in every convolution step, and FSGDN$_*$ denotes an FSDGN model without adding the term $\alpha\mathbf{I}$ in each convolution layer.

To begin with, the effect of initial connection is evaluated by FSDGN$_{\text{res}}$, which can be formulated by

$$\text{FSDGN}_{\text{res}} = \sigma\left(\left((1-\alpha)\widetilde{\mathbf{B}} \star \Theta_l + \alpha\mathbf{I}\right)\mathbf{H}^{l-1}\right). \tag{28}$$

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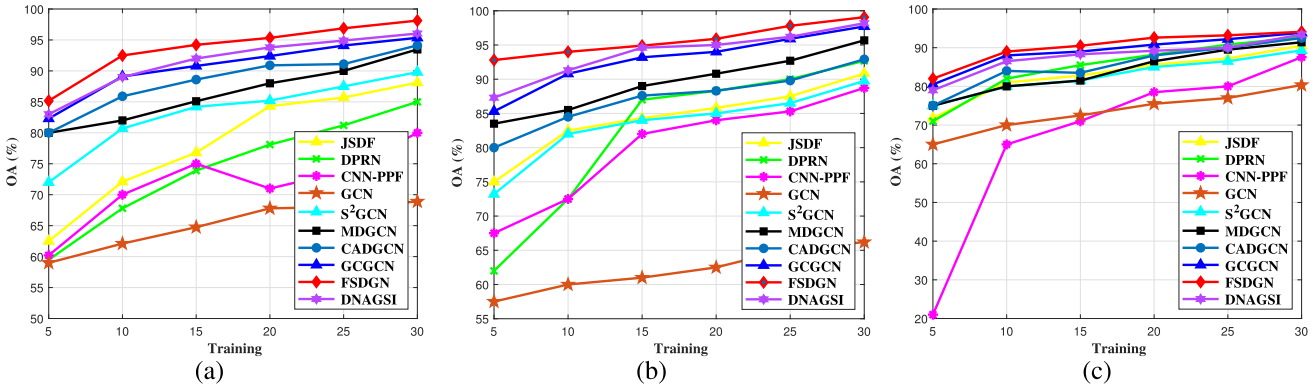

Fig. 9. OA of FSDGN with different numbers of training samples. (a) Indian Pines. (b) PaviaU. (c) Houston.

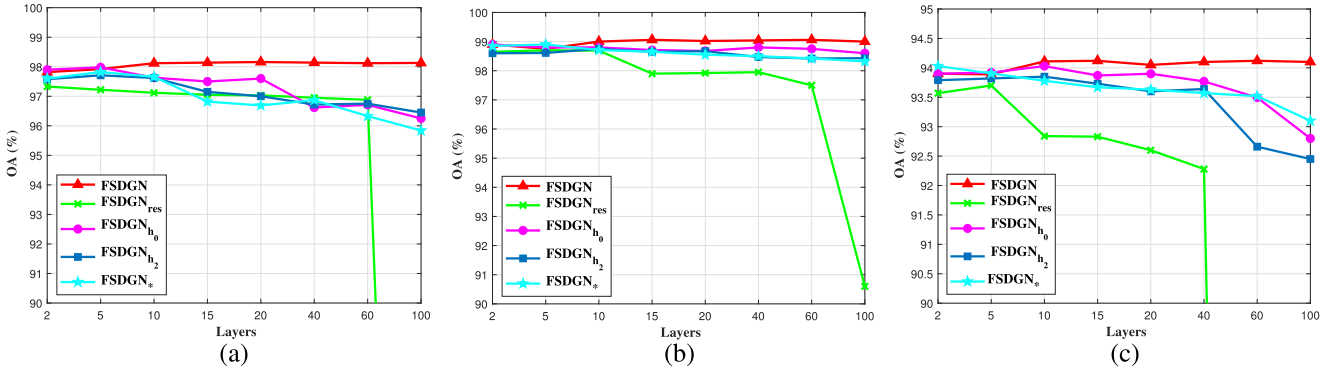

Fig. 10. OA of FSDGN, $\text{FSDGN}_{\text{res}}$, $\text{FSDGN}_{\text{h}_0}$, $\text{FSDGN}_{\text{h}_2}$, and FSDGN_* with different layers on three datasets. (a) Indian Pines. (b) PaviaU. (c) Houston.

We also evaluate the contribution of the addition of $\mathbf{H}^1$ by $\text{FSDGN}_{\text{h}_0}$, which is formulated as

$$\text{FSDGN}_{\text{h}_0} = \sigma\Big(\Big((1-\alpha)\widetilde{\mathbf{B}} \star \Theta_l + \alpha\mathbf{I}\Big)(1-\beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0\Big). \tag{29}$$

Moreover, we designed $\text{FSGDN}_{\text{h}_2}$ that introduces extra information from $\mathbf{H}^2$ in FSDGN, which is defined as

$$\text{FSDGN}_{\text{h}_2} = \sigma\Big(\Big((1-\alpha)\widetilde{\mathbf{B}} \star \Theta_l + \alpha\mathbf{I}\Big) \Big((1-\beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0\Big) + \mathbf{H}^2\Big). \tag{30}$$

Finally, we evaluate the effects of adding the term $\alpha\mathbf{I}$ for alleviating the oversmoothing problem by FSGDN_*, which is formulated as

$$\text{FSDGN}_* = \sigma\Big(\widetilde{\mathbf{B}}\big((1-\beta)\mathbf{H}^{l-1} + \beta\mathbf{H}^0\big)\Big)\Theta_l. \tag{31}$$

The results are given in Fig. 10. As shown in Fig. 10, FSDGN exceeds other components on every dataset and at most layers. Concretely, the performance of $\text{FSDGN}_{\text{res}}$ drops the fastest among these four models, which can be attributed to the absence of an initial feature connection operation, as we have discussed in Section IV. This phenomenon also proves that a partial self-loop measure is helpful to a certain degree but is not effective enough, especially when the model is very deep. Meanwhile, $\text{FSDGN}_{\text{res}}$ is more susceptible to the oversmoothing problem on Indian Pines than on other datasets. This is because the size of Indian Pines is smaller, which means that it has a much denser graph structure, and it is thus easier to be oversmoothed. The performance of $\text{FSGDN}_{\text{h}_0}$ and $\text{FSGDN}_{\text{h}_2}$ is relatively stable compared with $\text{FSDGN}_{\text{res}}$,

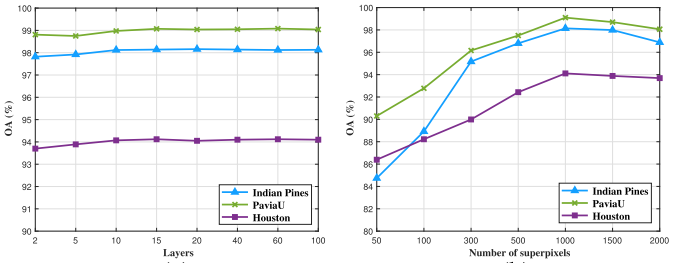

Fig. 11. (a) OA of FSDGN with different layers. (b) OA of FSDGN with different numbers of superpixels on three datasets.

indicating that the oversmoothing problem is ameliorated in the first two models. However, $\text{FSGDN}_{\text{h}_0}$ has a lower classification accuracy than FSDGN. A probable reason for this is the inaccuracy of the initial feature caused by the spectral variation. Moreover, $\text{FSGDN}_{\text{h}_2}$'s poor performance could be attributed to information redundancy, which is detrimental to obtaining discriminative graph representation. Besides, the performance of FSDGN_* is a little worse than FSDGN. This result shows that adding the method of partial self-loop can help alleviate the problem of oversmoothing in some extent, which is in accordance with the conclusion in Section IV.

F. Impact of Model Depth

To further verify the ability of FSDGN to address the oversmoothing problem, experiments were performed on three datasets, and the OA of FSDGN with different numbers of convolution layers was evaluated; the results are shown in Fig. 11(a). From the results, it is worth noting that

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TABLE V
CLASSIFICATION OA ON THREE DATASETS WITH THE CHANGE IN π

π	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Indian Pines	97.97	98.10	98.12	**98.14**	97.79	97.66	97.68	97.63	97.58	97.56	97.21
PaviaU	98.88	98.96	**99.05**	**99.05**	99.00	98.93	98.72	98.82	98.77	98.84	98.87
Houston	93.74	93.88	**94.11**	94.03	93.96	93.81	93.74	93.78	93.70	93.71	93.65

TABLE VI
CLASSIFICATION OA ON THREE DATASETS WITH THE CHANGE IN α

α	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Indian Pines	97.55	97.88	97.92	**98.14**	97.99	97.87	97.65	97.68	97.48	96.99	67.85
PaviaU	98.44	98.56	98.88	**99.05**	98.92	98.71	98.63	98.51	98.01	97.38	75.49
Houston	93.39	93.62	**94.11**	94.05	93.92	93.89	93.50	93.37	93.25	92.95	80.28

TABLE VII
CLASSIFICATION OA ON THREE DATASETS WITH THE CHANGE IN β

β	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
Indian Pines	96.25	97.73	98.11	**98.14**	98.05	97.91	97.82	97.70	97.56	97.35	96.80
PaviaU	98.12	98.77	**99.05**	98.97	98.81	98.80	98.74	98.62	98.55	98.49	97.86
Houston	93.28	93.79	93.95	**94.11**	94.01	93.75	93.72	93.68	93.65	93.45	93.22

the accuracy of FSDGN initially increases with the number of layers, demonstrating that developing deep models and extracting high-order semantic information are conducive to achieving more accurate classification results, as discussed. Then its accuracy stabilizes and remains at its highest level. This performance is totally different from that of previous GCN-based HSI classification models, whose performance noticeably degenerates when they are deeper than two layers. In contrast, our proposed FSDGN remains effective even with a hundred layers. All these observations prove that FSDGN is capable of addressing oversmoothing problems within the required depth, since the classification accuracy has become stable, and thus model with a depth exceeding that required for our experiments is therefore irrelevant.

G. Impact of Superpixel Segmentation

The experimental results of how the number of superpixels generated by the SLIC algorithm affects the model accuracy are shown in Fig. 11(b). It can be noted that on all the three datasets, the classification accuracy increases with the number of superpixels at first, and the model performs the best when the number of superpixels is 1000. But as the number of superpixels increases further, the performance of the FSDGN gets worse. Therefore, the number of superpixels is set to be 1000 for three datasets.

H. Hyperparameter Analysis

Quite a few hyperparameters have participated in our proposed FSDGN, and most of them are essential. To give a comprehensive analysis of the effect of changing their value, we investigated hyperparameters including π, α, and β and reported the experimental results in Tables V–VII.

1) *Influence of* π: π is the parameter that balances the contributions of pixel and superpixel topological relationship. Table V shows the OA of FSDGN for Indian Pines, PaviaU, and Houston 2013, with π varying between [0,1]. We can note that the classification accuracy turns out to be better when π ranges from 0.2 to 0.3. As a result, π was set as 0.3 in other experiments in this article.
2) *Influence of* α: α is the parameter that controls the proportion of a node's own feature and aggregated features from its neighborhood. Table VI reports the OA of FSDGN with α varying between [0,1], with an interval of 0.1, on three datasets. When α is 0, it means no self-loop is added, and the accuracy drops obviously. Meanwhile, if $\alpha = 1$, the model can be considered as a one-layer MLP no matter how many layers are stacked, which explains why there is a dramatic decrease in accuracy. According to Table VI, FSDGN performs better when α is approximately 0.2 or 0.3, so 0.3 is chosen as the value of α in Indian Pines and PaviaU experiments, and 0.2 is chosen as the value of $alpha$ in Houston experiments.
3) *Influence of* β: β is the parameter that controls the proportion of shallow feature information and deep feature information in graph representation. Table VII details the OA of FSDGN with β ranging from [0,1]. The model has the highest accuracy when β is about 0.2 or 0.3, and therefore β was set to be 0.2 in PaviaU and was set to be 0.3 in the other two datasets. In addition, the performance of FSDGN is worse when $\beta = 0$, demonstrating the importance of initial feature connection. When $\beta = 1$, the model only conducts feature aggregation on the initial graph, no matter how many layers are stacked. The model performance is certainly not satisfying because of the lack of feature propagation between distant nodes.

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VI. Conclusion

In this study, we propose an FSDGN. The proposed FSDGN differs from previous models in that it constructs a local and global topologically consistent graph for feature learning, and a novel deep graph convolution framework is designed for HSI classification to learn deep graph embedding. Meanwhile, comprehensive experiments are performed using three public HSI datasets, and the experimental results demonstrate the efficacy of FSDGN.

Despite achieving a state-of-the-art level of performance, there is still room for improvement. First, the training of FSDGN may be time-consuming because many layers are applied and the full batch gradient descent is performed. It is worth trying to address this issue using mini-batch training. In addition, the hyperparameters in FSDGN are artificially designed, so a method for automatically choosing parameters needs to be explored.

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